GND P2

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1 Postwar Environmental Changes in Japan

1.1 Overview

In the years following the end of the 2nd World War, the period was characterized by major industrial and urban development in Japan without regard for environmental protection. Industrial pollution and the resulting public health problems were largely ignored in pursuit of rapid and sustained industrial development. The early postwar period was marked by the single-minded concentration of the national government on economic reconstruction and catching up with the West. With the rapid economic growth of the 1960s and 1970s, industrialization caused heavy air and water pollution, and unchecked development destroyed the natural environment.

1.2 Reasons

There are three basic reasons for the intensity of environmental destruction in Japan. First, commercial enterprises based company decisions on economic growth and disregarded social responsibility. This was made possible partly because the Japanese government nurtured industrial capital, and partly because Japanese society values the organization above the individual. With a state-defined national goal of rapid industrialization, ethical considerations to restrain expanding corporate entities were meager. Second, the symbiotic relationship between the Japanese political system and government and business organizations exacerbated finding solutions to environmental problems. Third, the introduction of modern technologies used for land reclamation and other development projects frequently aggravated environmental pollution.

1.3 Industrial Concentration and Urbanization

In particular, exceptional congestion arising from the location of heavily concentrated industrial activities in crowded urban areas led to some of the most acute environmental pollution anywhere in the world. Over eighty percent of Japan's industrial production and most of the population are concentrated in a narrow zone stretching from the Kanto plain to northern Kyushu. This zone,

termed the Japanese Megalopolis, contains the three large industrial regions of Keihin, Hanshin, and Chukyo, which center respectively on the cities of Tokyo, Osaka, and Nagoya. Southwestward from Hanshin, this zone includes large industrial nuclei in Hiroshima and Kita Kyushu. Within each major industrial region, rapid industrial expansion was accompanied by fast population growth. The mounting geographical concentration of pollution-producing industry and the growing population was widely recognized as one of the major causes of serious environmental problems, ranging from air and water pollution to soil contamination.

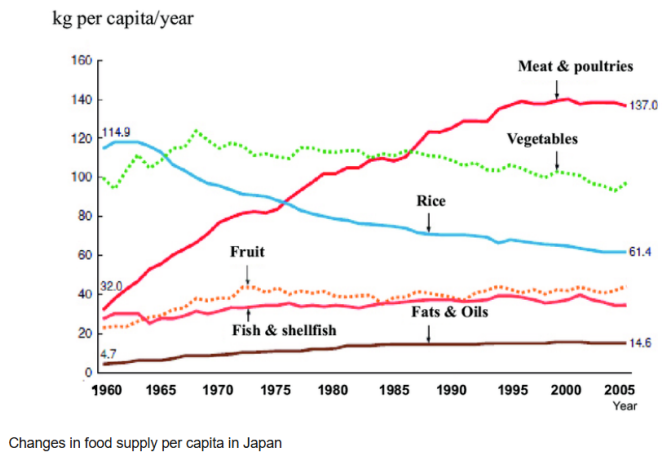

Figure 1: Changes in food supply per capita in Japan during rapid economic growth.

The rapid economic expansion significantly altered Japanese consumption patterns. As shown in Figure, the period between 1960 and 1980 saw a dramatic shift from traditional rice-based diets to meat and poultry. However, the continued high reliance on fish and shellfish meant that industrial water pollution, such as the mercury contamination in Minamata Bay, directly threatened a critical and stable component of the national food supply.

1.4 Dust and Air Pollution

During the first phase of high economic growth in the 1950s, the greatest environmental pollution problem was caused by dust and other airborne particulate matter. The main source of energy at that time was coal. Dustcollectors and other methods of particulate-matter control were either not provided or not working, and all of the chimneys belched forth black smoke. This situation continued into the next decade, so that by 1961 a major iron and steel complex in Yahata, northern Kyushu, was pouring 27 tons of particulate matter per day into the city's air, and in Kawasaki City, situated in the Tokyo Bay industrial

area, the amount was 23 tons. Along with the black smoke there was also a great amount of red smoke that spread over the sky.

The skies above Kawasaki

After the close of the Second World War, production technologies changed and the oxygen blast furnace was introduced. This made it possible to produce a better quality steel, but the reaction by-products included a great deal of particulate iron oxide, which was scattered far and wide. Soon after that the LD oxygen blast furnace was invented in Austria, and this produced even better quality steel by blowing oxygen into the furnace. However, this method also caused the release of a large amount of red smoke and iron oxide particulates, but Japanese industrial applications of the furnace did not include the use of dust-collectors. Because of this the amount of pollution from red smoke particulate matter greatly increased.

Many industrial workers were exposed to the industrial dusts in the atmosphere, and as a result developed various lung disorders. These problems were found most widely among mine workers and those who were involved in tunnel construction. In 1955, the government was forced to devise special protective measures in order to protect the miners from such occupational hazards. In the same year the Tokyo Metropolitan Government issued orders that made it mandatory to control heavy smoke and dust production. This was the first such ordinance related to airborne particulate matter since the end of the Second World War.

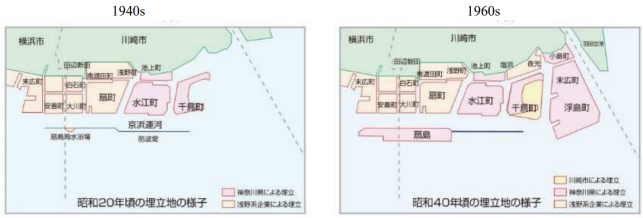

Figure 2: Transformation of the Kawasaki waterfront due to industrial land reclamation from the 1940s to the 1960s.

As shown in figure, rapid industrial expansion in Kawasaki involved extensive land reclamation, creating dense industrial zones that contributed significantly to severe air pollution.

1.5 Shift to Oil and Sulphur Pollution

In the 1960s, basic energy source changes from coal to heavy oil were instituted. As a result, the heavy smoke of coal combustion was replaced with invisible stack gases, but pollution problems were not thereby solved. Like coal, oil also contains sulphur which, when burned, results in sulphurous acid gases being discharged into the atmosphere. When coal was the main energy source, sulphurous acid gases were also a problem, but with the use of heavy oil in an expanding industrial economy, the problem became very serious.

A one-million-kilowatt, oil-fired, electric-power generation plant uses 1,300,000 tons of heavy oil per year. Since this fuel contains on average 2 per cent natural sulphur, 52,000 tons of sulphurous acid gases are released into the air annually from the power plant. Oil refineries and petrochemical complexes also release great amounts of such gases.

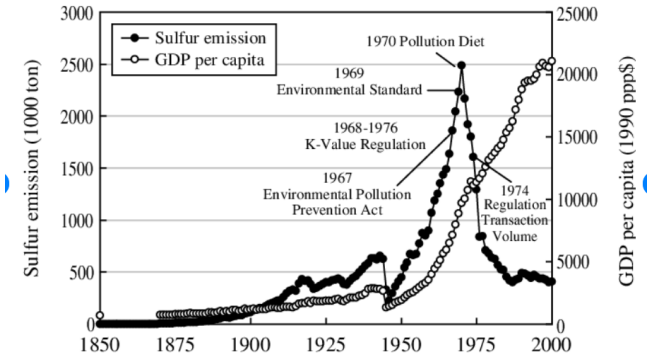

Figure 3: Correlation between GDP per capita and Sulfur Emissions in Japan (1850–2000).

1.6 Health Impacts: Yokkaichi Asthma

The fact that a great many patients developed respiratory diseases in areas where these complexes were located, especially along the Pacific Ocean coastline, became a very great social problem in the 1960s. The Yokkaichi petrochemical complex came into being in 1959 along the coast of Ise Bay, and in just one or two years respiratory diseases in the general population became painfully noticeable.

1.7 Water Pollution and Industrial Waste

Sulphurous acid gases are invisible and take their toll in silence, but black smoke can easily be seen and therefore everybody recognizes it as a health hazard. This fact is symbolic of the characteristics of post-Second World War environmental problems. The effluents from sulphuric-acid-based pulp-processing industries are the same in this regard, that is, they are not directly and immediately apparent as the cause of environmental problems.

A rayon-from-pulp textile production facility uses 38 per cent of the raw material by weight for the end-product, but 62 per cent of the post-process pulp is discharged as an effluent into the environment, usually without the application of pollution-prevention measures. If annual production levels reach 100,000 tons, over 160,000 tons of waste are produced. These red and black wastes discharged from the pulp-using production units found their way into natual waterways such as rivers and then into ocean systems. In this manner rivers and ocean estuaries were visibly polluted, and this destroyed fish resources, thereby compromising the viability of the fishing industry.

1.8 Chemical Pollution and Diseases

In the 1950s industrial mercury poisoning was found in Japan. The most famous cases were in the towns of Minamata and Niigata (Schreurs 36). It began when people who ate seafood from Minamata Bay began getting sick. The disease, known as Minamata disease, was caused by methyl mercury poisoning. Minamata disease affects the central nervous system. The symptoms include physical disabilities, coordination issues, trembles, and disfigurement. The disease can be deadly especially among babies. The Chisso Corporation dumped their waste, which contained the methyl mercury, into the Shiranui Sea. First fish, shellfish and seabirds began to die off. By 1953 the disease was evident in humans. This disease was also found in Niigata in 1965 (Karan 30).

Another deadly disease came about as the result of cadmium poisoning. This disease became known as itai-itai which translates as "it hurts" (Gordon 283). This disease causes brittle bones. It was discovered in 1955 in villages near the Jinzu River. In 1967 doctors were able to prove itai-itai disease was caused by wastes from the zinc refining plant of the Kamioka mines. Most victims of this disease were women and most cases resulted in death (Karan 31).

In 1968, PCB (polychlorinated biphenyl) poisoning was discovered in an area in and around Kitakyushu. The Japanese government acted too late when it halted PCB production and limited the uses of mercury to certain closed-system processes.

1.9 Land Reclamation and Ecological Damage

Another characteristic of Japan's environmental destruction during the post-war period is the concentration in limited land areas of manufacturing plants much larger in scale than similar factories found in Western countries.

Along the coastline of Tokyo Bay, land reclamation has taken place at such a rapid pace that the original coastline is no longer discernible.

In the seaweed-producing areas along the Inland Sea coast, huge amounts of concrete were brought to unprecedentedly large construction projects, resulting in massive destruction of the biosphere.

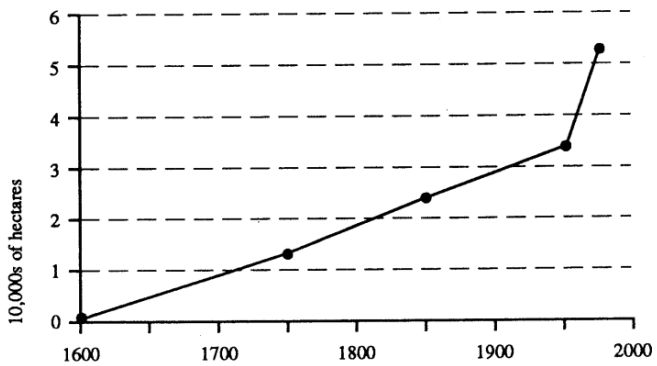

Fig. 2.1. Inland Sea Land Reclamation Expansion (after Yoshiro Haraguchi, "Seto-naikai ni okeru umetateno genjo to mondaiten," *Ao to midori* (September 1974).

Figure 4: Expansion of land reclamation in Japan's Inland Sea over time.

As shown in figure, land reclamation expanded rapidly during periods of industrial growth, contributing significantly to ecological degradation in coastal regions.

1.10 Health Consequences and Victims

Once the health of the human body has been compromised by the polluted environment, it is exceedingly difficult to return bodily functions to normal levels again.

The number of government-recognized pollution victims found in the most severely poisoned areas stood at 81,222 persons as of 1980.

These patterns of rising environmental degradation followed by later improvements can be better understood through an economic perspective.

1.11 Environmental Kuznets Curve (EKC) Interpretation

Japan's environmental history after World War II follows the idea of the Environmental Kuznets Curve (EKC). This idea says that as a country grows economically, pollution first increases, but after reaching a certain level of income, pollution starts to decrease.

In Japan, from the 1950s to the early 1970s, pollution increased rapidly. This was because of fast industrial growth, crowoded cities, and environmental laws which were weak. During this time, there were high levels of air and water pollution, along with serious health problems like Minamata disease.

After the 1970s, the situation started improving. People became more aware of environmental issues, the government introduced stricter laws, and new technologies helped reduce pollution. Even though the economy kept growing, pollution levels started to fall.

So, Japan is a strong real-world example of how the Environmental Kuznets Curve works.

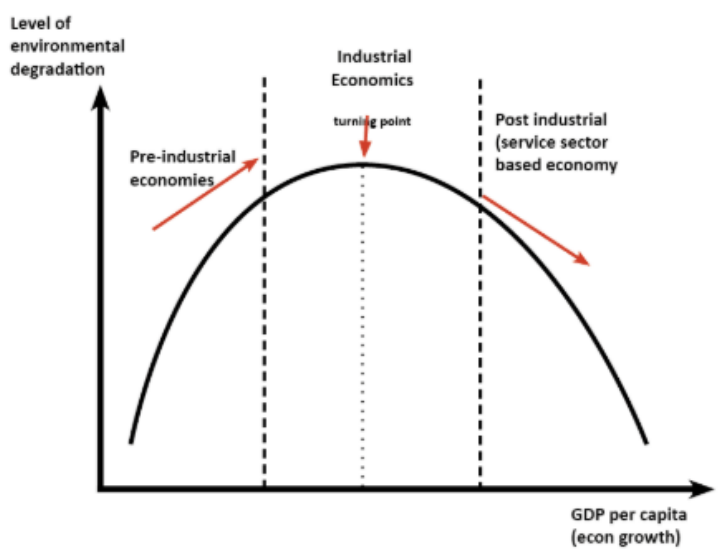

Figure 5: Environmental Kuznets Curve (EKC) illustrating the relationship between economic growth and environmental degradation.

2 References

- Schreurs, M. A. (2002). *Environmental Politics in Japan, Germany, and the United States.*
- Karan, P. P. (2005). *Japan in the 21st Century.*
- Gordon, A. (2003). *A Modern History of Japan.*